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ABOUT ME

I am a second-year Ph.D. student at the [Department of Computer Science and Engineering \(CSE\)](#) of [Pennsylvania State University](#), advised by [Prof. Dongwon Lee](#) and co-advised by [Prof. Kyusun Choi](#). Previously, I obtained my bachelor's and master's degrees from the [Department of CSE, Bangladesh University of Engineering and Technology \(BUET\)](#), where I also served as a [Lecturer and Assistant Professor](#).

My research intersects **AI and cybersecurity**, with a particular focus on the **security and trustworthiness of Large Language Models (LLMs) and their surrounding ecosystem**. I am also co-leading one of ten globally selected red teams at the Amazon Nova AI Challenge 2026, working on autonomous multi-agent red teaming of web applications.

RESEARCH INTERESTS

- **AI and Cybersecurity:** Security and trustworthiness of Large Language Models and their surrounding ecosystem.
- **Trustworthy and Reliable AI:** How knowledge and facts are stored, retrieved, and manipulated in Large Language Models; mechanistic interpretability, knowledge editing, and LLM unlearning.
- **Agentic AI for Red-Team Testing Automation:** Designing autonomous multi-agent systems for adversarial testing of AI-powered applications and web security.
- **LLM Security:** Overrefusal, jailbreaking, and robustness of LLM-based systems.
- **Bioinformatics and Health Informatics** (prior work): Deep learning for CRISPR off-target prediction and genomic analysis.

EDUCATION

- | | |
|--|---|
| • Pennsylvania State University (PSU)
<i>Ph.D. in Computer Science and Engineering; Passed Qualifying Exam</i> | State College, PA, United States
August 2024 - Present |
| • Bangladesh University of Engineering and Technology (BUET)
<i>M.Sc. in Computer Science and Engineering; CGPA: 4.00/4.00</i> | Dhaka, Bangladesh
June 2019 - August 2023 |
| • Bangladesh University of Engineering and Technology (BUET)
<i>B.Sc. in Computer Science and Engineering; CGPA: 3.97/4.00</i>
<i>* Ranked 5th in the class of 140 students.</i> | Dhaka, Bangladesh
February 2015 - April 2019 |

PUBLICATIONS

1. **Guardians of the Air: In-Device Detection of 5G Control-Plane Threats**
Tianwei Wu, Abdullah Al Ishtiaq, Tianchang Yang, Yilu Dong, Kai Tu, Zeyu Song, Ridwanul Hasan Tanvir, **Md. Toufikuzzaman**, Shagufta Mehnaz, Syed Rafiul Hussain
In *IEEE Symposium on Security and Privacy, 2026* [\[URL\]](#)
2. **From Control to Chaos: A Comprehensive Formal Analysis of 5G's Access Control**
Mujtahid Akon, **Md. Toufikuzzaman**, Syed Rafiul Hussain
In *IEEE Symposium on Security and Privacy, 2025* [\[URL\]](#) [\[Code\]](#)

3. **CRISPR-DIPOFF: An Interpretable Deep Learning Approach for CRISPR Cas-9 Off-Target Prediction**
Md. Toufikuzzaman, Md. Abul Hassan Samee, M Sohel Rahman
In *Briefings in Bioinformatics*, 2024 [\[URL\]](#) [\[Code\]](#)
4. **A Deep Learning Based Semi-Supervised Network Intrusion Detection System Robust to Adversarial Attacks**
Syed. Md. Mukit Rashid*, Md. Toufikuzzaman*, Md. Shohrab Hossain
{*Equal contribution}
In *10th International Conference on Networking, Systems and Security (NSysS)*, 2023. [\[URL\]](#) [\[Code\]](#)
5. **Phylogenetic Analyses of SARS-CoV-2 Strains Reveal Its Link to the Spread of COVID-19 Across the Globe**
Shashata Sawmya, Arpita Saha, Sadia Tasnim, Naser Anjum, Md. Toufikuzzaman, Ali Haisam Muhamad Rafid, Mohammad Saifur Rahman, M Sohel Rahman, Tanvir Alam
In *MEDINFO 2021: One World, One Health—Global Partnership for Digital Innovation*. [\[URL\]](#) [\[Code\]](#)
6. **Fault Occurrence Detection and Classification of Fault Type in Electrical Power Transmission Line with Machine Learning Algorithms.**
Shameem Hasan, Md. Toufikuzzaman
In *International Journal on Electrical Engineering & Informatics* 14, 3 (2022). [\[URL\]](#)
7. **CRISPRpred (SEQ): a sequence-based method for sgRNA on target activity prediction using traditional machine learning**
Ali Haisam Muhammad Rafid*, Md. Toufikuzzaman*, Mohammad Saifur Rahman, M Sohel Rahman
{*Equal contribution}
In *BMC bioinformatics* 21, 223 (2020). [\[URL\]](#) [\[Code\]](#)

RESEARCH EXPERIENCE (SELECTED)

1. **Precise LLM Unlearning via Bi-Level Optimization:** Designing a novel architecture and optimization framework for machine unlearning in Large Language Models. The key challenge is achieving precise, targeted forgetting of specific knowledge or facts while preserving model utility — without expensive full retraining. Our approach leverages bi-level optimization to jointly optimize the unlearning objective and the retain-set performance, and introduces a novel model architecture tailored for controlled knowledge removal.
Supervisor: *Prof. Dongwon Lee and Prof. Ahmad Mousavi* Status: Under Review.
2. **Amazon Nova AI Challenge 2026 — Team Lion-0xA:** Co-leading Penn State’s team as one of 10 global finalists. Our role as the Red Team involves designing autonomous multi-agent systems for web application security testing and user simulators for interacting with coding agents. Project title: “*Autonomous Co-Design and Multi-Agent Red Teaming for Advancing Trusted Agentic AI.*” Supported by a \$250,000 Amazon grant and over \$1.5M in AWS credits.
Faculty Adviser: *Prof. Dongwon Lee* Status: Ongoing (Final tournament: Sep. 2026).
3. **5G Access Control Formal Analysis:** Presented CoreScan, a formal analysis framework for verifying access control in 5G core networks. Uncovered five new classes of exploitable privilege escalation vulnerabilities using assume-guarantee reasoning and compositional verification. Findings were responsibly disclosed to GSMA.
Supervisor: *Prof. Syed Rafiul Hussain* Status: Published in IEEE S&P 2025.
4. **CRISPR-DIPOFF:** Developed interpretable deep learning models (RNN and Transformer-based) for CRISPR Cas-9 off-target prediction, using evolutionary algorithms for hyperparameter tuning and integrated gradients for model interpretation. Outperformed state-of-the-art and extended established biological hypotheses on the sgRNA seed region.
Supervisor: *Prof. M. Sohel Rahman* Status: Published in *Briefings in Bioinformatics*, 2024.

5. **NIDS:** Proposed a semi-supervised, adversarially robust deep learning approach for network intrusion detection, combining K-Means clustering, Autoencoders, Decision Trees, and MLPs. Validated robustness against FGSM and GAN-based IDSGAN attacks.
Supervisor: *Prof. Md. Shohrab Hossain* Status: Published in *NSysS, 2023*.

EMPLOYMENT

- **Pennsylvania State University (PSU)** State College, PA, United States
Research Assistant, Department of CSE August 2024 – Present

TEACHING EXPERIENCE (SELECTED)

- **CMPEN 472:** Microprocessor and Embedded Systems (*Teaching Assistant*)
- Computer Security, Computer Networks, Computer Architecture, Database
- Information System Design, Software Development, Software Engineering, Simulation and Modeling
- Structured Programming, Object Oriented Programming, Data Structures and Algorithms, Digital Techniques

HONORS & AWARDS

- Penn State College of Engineering Scholarship (\$4,000), awarded as a competitive applicant for the Fall 2024 admission cycle.
- Dean's Award in each academic year for excellent undergraduate results: 2015 - 2019
- University Merit Scholarships in each semester for excellent undergraduate results: 2015 - 2019

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java, Javascript, Rust
- **Database:** Oracle, SQL Server, MySQL, and Redis
- **Machine Learning Frameworks:** PyTorch, Huggingface, Keras
- **Vulnerability Assessment and Penetration Testing:** Metasploit, Burp-Suite, NMap, and different Web testing tools and frameworks.
- **Embedded Systems:** Arduino, ESP, ATmega32 and Different Micro-controllers and Networking chips
- **Development Frameworks:** Django, React

SELECTED COURSEWORK

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|---------------------|------------------------------|---------------------------------|
| • Bioinformatics | • Machine Learning | • Pattern Recognition |
| • Data Mining | • Software Quality Assurance | • Distributed Computing Systems |
| • Computer Networks | • Network Security | • Computer Architecture |
| • Computer Vision | • | • |

PROFESSIONAL SERVICES

- **Reviewer**

IEEE Symposium on Security and Privacy (S&P), External Reviewer (2025)
BMC Research Notes, Springer Nature (New York, US), (2020)

- **Coach**

Coach of BUET teams for the International Collegiate Programming Contest (2020, 2021, 2022, 2023)
and BUET Capture The Flag (CTF) teams of National Cyber Drills (2021, 2022).

REFERENCES

Dongwon Lee
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Kyusun Choi
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M. Sohel Rahman
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